

#Day 8: Database

Database

- A Database is used to store, organize, and retrieve application data.
- It can store structured data (tables) and unstructured data (documents, images, etc.).

Common Databases

- MySQL – Relational SQL database.
- PostgreSQL – Advanced relational database.
- MongoDB – NoSQL document database.
- Supabase – Cloud-based PostgreSQL service.

Database Schema

- A Schema defines the structure of a database.
- It specifies tables, columns, data types, relationships, and constraints.

SQL Database (MySQL)

- Stores data in tables consisting of rows and columns.
- Each row represents a record, and each column represents an attribute.

Example: Users Table

ID	Username	User Type	Email
123	Jacob	Admin	jacob@gmail.com
124	Joseph	User	joseph@gmail.com

MongoDB (NoSQL)

- MongoDB stores data as collections containing documents.
- Documents are stored in JSON/BSON format, making it flexible for changing data.

Example

```
{
  "id": 123,
  "username": "Jacob",
  "userType": "Admin",
  "email": "jacob@gmail.com"
}
```

Server

- A Server is a computer that hosts applications and databases.
- It processes requests from users and sends responses.

- Typical server configuration:
 - 64 CPU Cores
 - 128 GB RAM
 - 16 TB Storage
- Runs continuously (24×7) with backup power and cooling.
-

Cloud Platforms

Cloud platforms provide computing resources over the internet.

- AWS – Amazon Web Services
- Azure – Microsoft Cloud Platform
- GCP – Google Cloud Platform

Benefits

- Easy deployment
- High availability
- Scalability
- Data backup

Container

- A Container packages an application with all its dependencies.
- It ensures the application runs the same on every system.
- Docker creates containers.
- Kubernetes manages multiple containers in a cluster.

Virtual Machine (VM)

- A Virtual Machine is a virtual computer running on a physical server.
- Each VM has its own Operating System (Windows/Linux) and resources like CPU, RAM, and storage.

MySQL Installation

Steps:

1. Install MySQL Server.
2. Create a database.
3. Create tables.
4. Connect the backend application to MySQL.

MongoDB Atlas

- MongoDB Atlas is the cloud version of MongoDB.
- It allows you to create and manage databases online.
- It provides a connection string to connect your application securely.

MongoDB Compass

- MongoDB Compass is the graphical user interface (GUI) for MongoDB.
- It is used to:
 - Connect to MongoDB Atlas.
 - View databases and collections.
 - Insert, update, delete, and search documents.